VR-Sets. Preprint.
Part II. The Operational Construction of Sets

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II.0. Introduction

This part of the preprint formalises the operational construction of sets. Building on the ontological position fixed in Part I (∅ as the sole primitive; everything else as operationality), we introduce four definitions forming the core of the system:

- **Definition 1 —** what a set is in VR-Sets.
- **Definition 2 —** what ∅ is.
- **Definition 3 —** three kinds of sets distinguished by the character of their operationality (finite, infinite, cyclic).
- **Definition 4 —** operational identity ≡.

From these four definitions three lemmas are derived (extensionality, uniqueness of ∅, operational depth); the closure principle is stated — the principal instrument by which the ZF axioms will be proved as theorems in Part III; and Russell's paradox is dissolved ontologically. The part closes with a set of characteristic examples.

The presentation is structured so that all of Parts III and IV rest only on this text; nothing from the later parts is presupposed. This part is the self-contained foundation.

II.1. Core definitions

Four definitions forming the core of the operational ontology of sets. They are fixed in their final form — changes here would affect all that follows.

Definition 1 (set)

A **set** in VR-Sets is an operational entity given by its membership functionality: upon a query, the functionality either reveals an element (which is itself a set, possibly the same set as the one being queried) or reveals nothing. The order in which elements are revealed is an artefact of observation and does not enter into the identity of the set. A set has no internal structure beyond its functionality; it *is* its functionality.

Commentary. (1) A set in VR-Sets is not a container with contents, but a description of behaviour: "here is what I will reveal when asked." This replacement of the underlying metaphor is the core divergence from classical ZF (see Part I §I.4).

(2) "A set is its functionality" is an ontological identity, not an operationalism by definition. The functionality is not "how the set behaves" (as though something more fundamental stood behind the behaviour); the functionality is the set. No "carrier of the functionality" distinct from it is presupposed.

(3) The elements of sets are themselves sets. This is explicitly fixed in the definition ("an element is itself a set") and accords with Position 2 from §I.3: in VR-Sets there are no atoms, individuals, or urelements.

(4) Self-reference ("the same set as the one being queried") is explicitly admitted in the definition. This is an operational possibility; restrictions on self-reference pertain to the choice of mode (Part IV), not to the definition of a set as such.

(5) The order of revelation is an artefact of observation. This means that if a functionality reveals elements a, b in one observed order and b, a in another, both observations relate to the same functionality. The identity of a set is determined by *which* elements are revealed, not *in what order*.

Definition 2 (∅)

∅ is the unique set whose functionality reveals nothing: ∅ is the empty operationality.

Commentary. (1) The definition appeals to "uniqueness," which will be proved as Lemma 2. Here uniqueness is fixed as part of the meaning of the symbol ∅: when we write ∅, we mean precisely this set, not one among many possible empty sets.

(2) ∅ is the sole object in the ontology of VR-Sets regarding which ontological primacy is asserted. All other sets are operationalities; ∅ is empty operationality, and this "empty" is the terminating point of every construction.

Definition 3 (kinds of sets by operationality)

A set A is:

- *finite* if its functionality exhausts all elements in a finite number of queries;
- *infinite* if its functionality is not exhausted at any finite number of queries;
- *cyclic* if among the revealed elements its functionality returns A itself.

These properties are not mutually exclusive.

Commentary. (1) Finiteness is defined through exhaustion: ∅ and all the von Neumann ordinals O_n are finite. Every element of any finite set has finite operational depth (Lemma 3).

(2) Infinity is defined through *non-exhaustion*, not through cardinality. Example: ω is infinite — its functionality reveals O_0, O_1, O_2, … without termination.

(3) Cyclicity is defined through self-return. The Quine atom A = {A} is cyclic. The set B = {∅, B} is cyclic (its functionality returns B among its other elements). Cyclic sets belong to the ZFA-mode but not the ZFC-mode.

(4) The properties are not mutually exclusive: a set may be simultaneously infinite and cyclic (for instance, a describable sequence of sets each containing a reference to the next and to itself). Such cases are rare in substantive mathematics but are not excluded by the definition.

Definition 4 (operational identity ≡)

Sets A and B are *operationally identical* (A ≡ B) if their functionalities coincide: for every element x of A there exists an element y of B with x ≡ y, and conversely. ∅ ≡ ∅.

Commentary. (1) The definition is coinductive: ≡ is defined through ≡ on elements. The termination point is ∅: if both sets are empty, ≡ holds trivially (vacuously).

(2) Relation to Leibnizian equality from VR. In VR (Part I, Definition 2) equality is given classically: $x = y \iff \forall p: p(x) \leftrightarrow p(y)$ — coincidence under all properties. In VR-Sets ≡ is the direct analogue of Leibniz, lifted to the operational level: in place of "under all properties" — "under all responses to queries." Substantively, this is the same identity relation, expressed in the language suited to the ontology.

(3) The distinction between ≡ and ordinary =. In what follows ≡ is used for the operational identity of sets, and = only in contexts referring to classical objects (ZF formulas, numerical equalities). This discipline is observed in all parts of the preprint. The sign ≡ is also distinguished from its various standard senses in mathematical literature (modular congruence, logical equivalence, definitional equality); see Part I §I.6 for the note on notation.

(4) Bisimulation. In the AFA literature (Aczel 1988) the analogous relation is known as bisimulation: sets are equivalent if their graphs of membership are isomorphic (allowing cycles). In VR-Sets ≡ is structurally the same relation, expressed operationally. The direct connection is discussed in Part IV §IV.8.

II.2. Lemmas

Three consequences embedded in the definitions. They are used as lemmas in the later parts.

Lemma 1 (extensionality)

If A and B have the same elements (up to ≡), then A ≡ B.

Proof. This is a direct reading of Definition 4: the condition of coincidence of functionalities in Definition 4 is precisely "for every element x of A there exists y of B with x ≡ y, and conversely." The statement "sets with the same elements are distinct" contradicts the definition of identity and is inexpressible in VR-Sets.

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Remark. This is not a theorem in the strict sense but a formal restatement of the definition. In classical ZF extensionality is a separate axiom, because in the classical approach equality is given independently of membership. In VR-Sets equality (more precisely, operational identity ≡) is defined *through* membership, so extensionality is built into the definition.

Lemma 2 (uniqueness of ∅)

There exists exactly one set with an empty functionality.

Proof. If two functionalities both reveal nothing, the coincidence condition of Definition 4 holds vacuously on both sides: "for every element x of A there exists y of B with x ≡ y" — the premise is false, the implication is true; and conversely. Therefore A ≡ B.

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Remark. Lemma 2 does not assert the existence of ∅ — this is given by Definition 2 as part of the ontology. The lemma asserts uniqueness: in writing ∅ we refer to a fully determined object, not to an arbitrary "empty set" among many.

Lemma 3 (operational depth)

Every set has an operational depth — the least number of unfolding steps after which ∅ is reached. The depth of ∅ is 0. Finite sets have finite depth. Infinite and cyclic sets do not have finite depth (for cyclic sets, unfolding does not terminate at ∅).

Note on the status of the lemma. Lemma 3 is not an existence theorem but a definition of a property of a functionality. Every set A *has* its depth: a number (or "infinity" with the qualification "non-cyclic") — a structural characteristic of A. For finite sets this number is finite; for ω this number is non-finite but each concrete unfolding terminates (see Part IV §IV.2); for cyclic sets unfolding does not terminate at all.

Significance of Lemma 3. This lemma provides the instrument of *induction on sets* in VR-Sets — the analogue of axiom A4 from VR, lifted to the level of sets. The numbers of VR (the von Neumann ordinals O_n) have operational depth exactly n; ω has non-finite depth but is non-cyclic — this is essential for distinguishing "infinite by enumeration" from "cyclic."

II.3. The closure principle

The principal instrument of VR-Sets. Through it all the ZF axioms are proved in Part III.

Statement

The closure principle of VR-Sets. If a functionality F is operationally describable — that is, if there exists an operational procedure that gives a definite response to every query (either revealing a specific element or revealing nothing) — then there exists a set A such that A is F.

A description requiring a functionality to give contradictory responses to one and the same query (as in the definition R = "reveal A if and only if A is not revealed") **does not** specify an operationally describable functionality, and does not give rise to a set.

Commentary

(1) The closure principle converts the axioms of set existence (as they appear in ZF) into *closure theorems* of the operational universe relative to specific describable constructions. There is no need to postulate "⋃A exists" — one must show that ⋃A is operationally describable, and then by the closure principle it is a set. This is a substantive simplification: in place of a long list of existence axioms, a single general principle.

(2) The closure principle is neither a definition nor a theorem, but an *ontological choice*: which functionalities count as sets. The choice is motivated by the slogan "only ∅ is, all else is doing": all that exists as describable action exists as a set; no different ontological status is presupposed for "potential" or "inaccessible" entities.

(3) "Operational describability" is a substantive notion. A description must: (a) be finite (a syntactic record in a formal language); (b) give a definite response to every query (not contradictory, not undefined, not undecidable by construction). When these conditions hold, the description specifies a functionality, and the closure principle applies.

(4) In the two modes of VR-Sets the closure principle is applied differently. In the ZFC-mode — only to functionalities whose successive unfoldings terminate at $\varnothing$. In the ZFA-mode — to all describable functionalities. Part IV unfolds this distinction.

Resolution of Russell's paradox

Russell's paradox is dissolved at the level of the definition. The description R = "reveal x if and only if $x \notin x$" demands a contradictory response from the functionality to the query "$R \in R$?":

- If the functionality R reveals R on this query, then by the description (R is to contain only x with $x \notin x$) we have $R \notin R$, so R *should not* reveal R. Contradiction.
- If R does not reveal R, then by the description R should contain R (since $R \notin R$). So R *should* reveal R. Contradiction.

This is a contradictory description. By the closure principle R **is not** a set — not because we forbade it by a separation schema (as in ZF) or by type stratification (as in type theory), but because such a functionality *does not exist* operationally. At each concrete step a functionality either reveals a specific element or does not; no third option exists; the contradictory definition of R cannot be carried out by any procedure.

Relation to classical approaches

Classical ZF resolves Russell's paradox by the **axiom of separation**: one cannot form a set from all x with an arbitrary property, only from $x \in A$ with a property. This is a syntactic restriction imposed on the schema of construction.

Type theory resolves the paradox by **stratification**: an element and a set must lie at different levels; "$x \in x$" is a type violation.

VR-Sets resolves the paradox by **operational definiteness**: paradoxical descriptions simply do not specify functionalities. The result is the same (no paradox), but the cause is ontological, not syntactic. This is consistent with the general programme of VR: ontological clarity is placed ahead of technical devices.

II.4. Principal examples

Specific operational sets illustrating the definitions and lemmas.

Example 1: $\varnothing$

The functionality $\varnothing$ reveals nothing. Operational depth — 0. Finite (exhausted in 0 queries), not infinite, not cyclic. Unique by Lemma 2.

Example 2: {∅}

The functionality {∅} reveals ∅, and on all subsequent queries reveals nothing. Operational depth — 1 (one unfolding step suffices to reach ∅). Finite.

In the terminology of VR this is O_1 — the first non-zero von Neumann ordinal, obtained by applying t to ∅: $O_1 = t(\varnothing) = \varnothing \cup \{\varnothing\} = \{\varnothing\}$.

Example 3: {∅, {∅}}

The functionality reveals ∅ and {∅}. Operational depth — 2. Finite.

In the terminology of VR this is $O_2 = t(O_1) = O_1 \cup \{O_1\} = \{\varnothing, \{\varnothing\}\}$.

Example 4: ω

The functionality ω reveals on its n-th query the von Neumann ordinal O_n: ∅, {∅}, {∅, {∅}}, …, without termination.

Operational depth — non-finite (unfolding does not reach ∅ at any finite step). Infinite (the functionality is not exhausted). **Non-cyclic**: ω does not return itself among the revealed elements. Each revealed element is a specific O_n with finite n; the unfolding chain from any specific O_n terminates at ∅ in n steps.

This last characteristic — "infinite but non-cyclic" — gives ω the property of well-foundedness, unfolded in Part IV §IV.2: every successive unfolding of ω terminates at ∅, even though ω itself has no finite operational depth.

Example 5: the Quine atom A = {A}

The functionality A reveals A on every query. Operational depth is undefined: the unfolding $A \ni A \ni A \ni \ldots$ does not terminate at ∅. Cyclic by Definition 3. Not well-founded.

The Quine atom belongs to the ZFA-mode of VR-Sets but not to the ZFC-mode. This is the first example of an ontologically legitimate but non-well-founded set; detailed discussion appears in Part IV §IV.7.

Example 6: the set B = {∅, B}

The functionality B reveals ∅ and B. Cyclic (reveals itself among its elements). The unfolding chain $B \ni \varnothing$ terminates at the first step; the chain $B \ni B \ni B \ni \ldots$ does not terminate. Since well-foundedness requires termination of **every** chain, B is not well-founded.

This example shows that cyclicity need not be trivial (as with the Quine atom): a set may contain both ordinary elements and a self-reference.

Example 7: a description that does not specify a set

R = "reveal x if and only if $x \notin x$" (Russell's definition as considered above). This description is contradictory — it requires the functionality to give a contradictory response to the query "$R \in R$?". R is not operationally describable. R is not a set.

This is a critically important example: it shows that not every syntactic record specifies a set. The closure principle yields "every describable functionality is a set," but *describable* is a substantive condition, not empty: contradictory descriptions are excluded.

II.5. Summary and transition to Part III

In this part we have introduced the formal apparatus on which the remainder of the work rests:

(1) **Four definitions**: of a set, of ∅, of kinds of sets by operationality, of operational identity ≡.

(2) **Three lemmas**: extensionality (≡ is built into Definition 4), uniqueness of ∅, operational depth.

(3) **The closure principle**: every operationally describable functionality is a set. This is the principal instrument of the constructions that follow.

(4) **Resolution of Russell's paradox** — ontologically, not syntactically: paradoxical descriptions do not specify functionalities.

(5) **Seven characteristic examples** covering the principal ontological situations: ∅, finite ordinals, ω, the Quine atom, a partially cyclic set, a contradictory description.

What follows. Part III applies this apparatus to the axioms of classical ZF. Each ZF axiom (except extensionality and the existence of ∅, already built into Part II) becomes a **closure theorem** of the operational universe — that is, an assertion that a corresponding describable functionality is a set under the closure principle.

Part IV formally separates the two modes in which the closure principle is applied — the ZFC-mode (only to well-founded functionalities) and the ZFA-mode (to all describable ones) — and establishes theorems of completeness for each mode relative to the corresponding classical theory. The Quine atom and its analogues are legitimate objects of the ZFA-mode but not of the ZFC-mode.